Context Diagram
UML Use Case Diagram

Example system: Library Management System

1. Context Diagram

Shows the WHOLE system as ONE shape, everything OUTSIDE it, and the DATA between them.

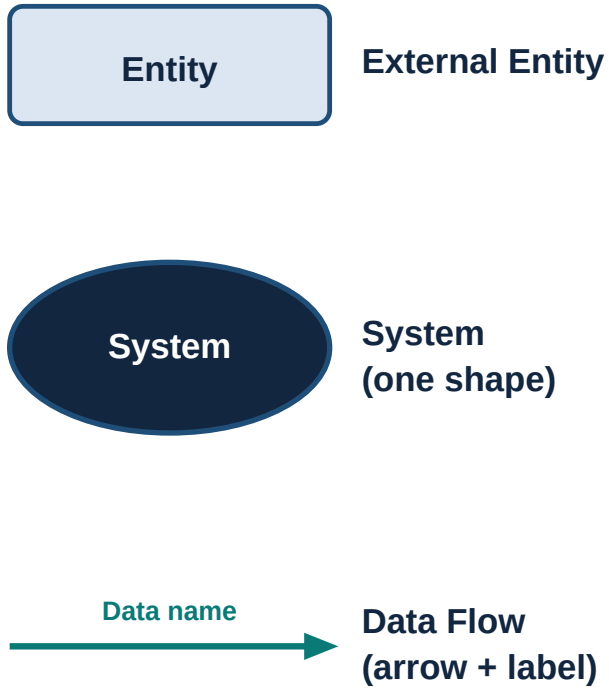
Key points

- ONE system shape (no inside details)
- External entities (people / other systems)
- Data flows (arrows labeled with data)

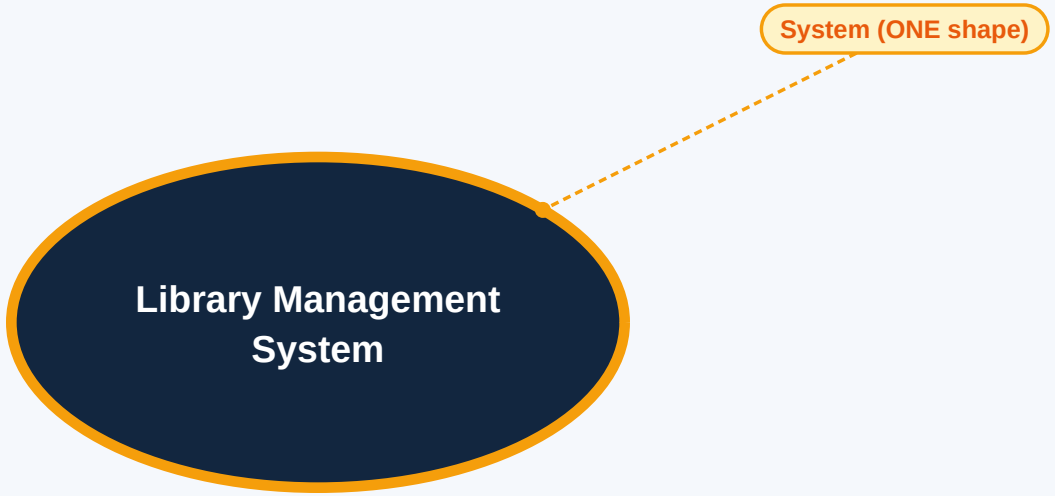
Rules

Every arrow touches the system.
Label arrows with data (nouns).

Symbols



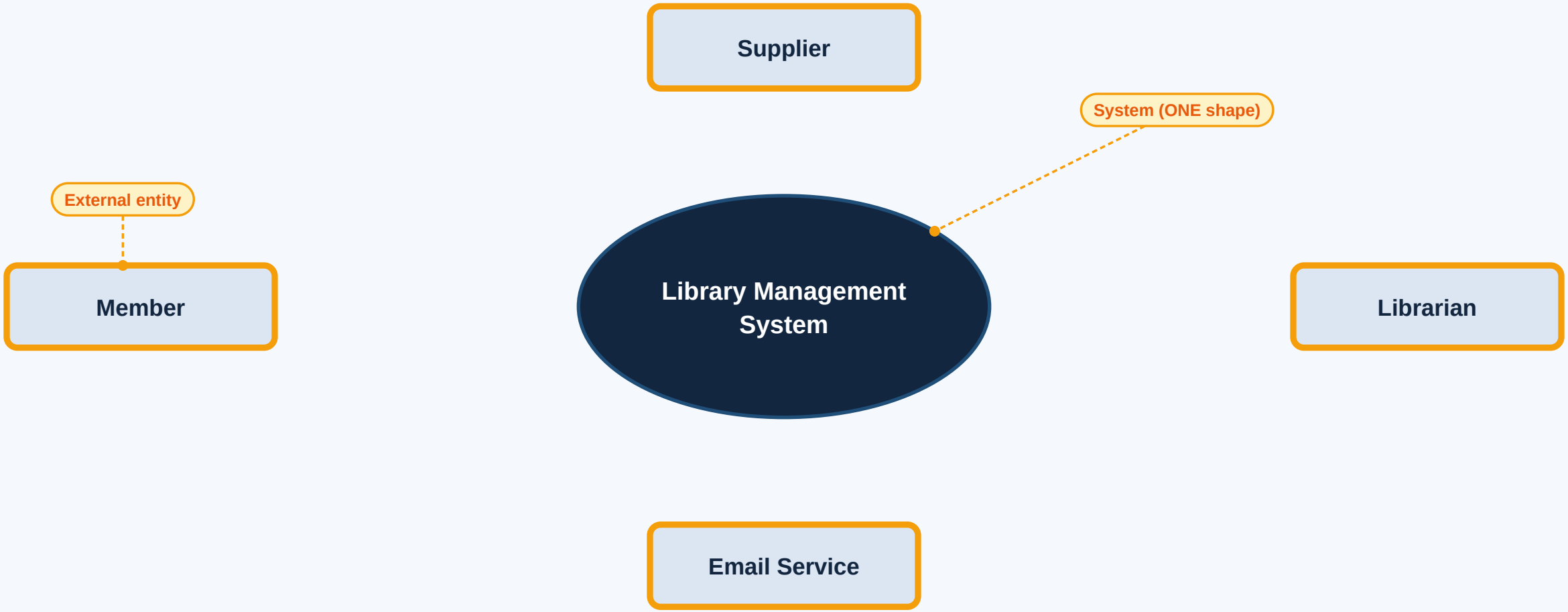
Building the Context Diagram



Step 1: Draw the system as ONE shape

Give it a name. Ignore what is inside.

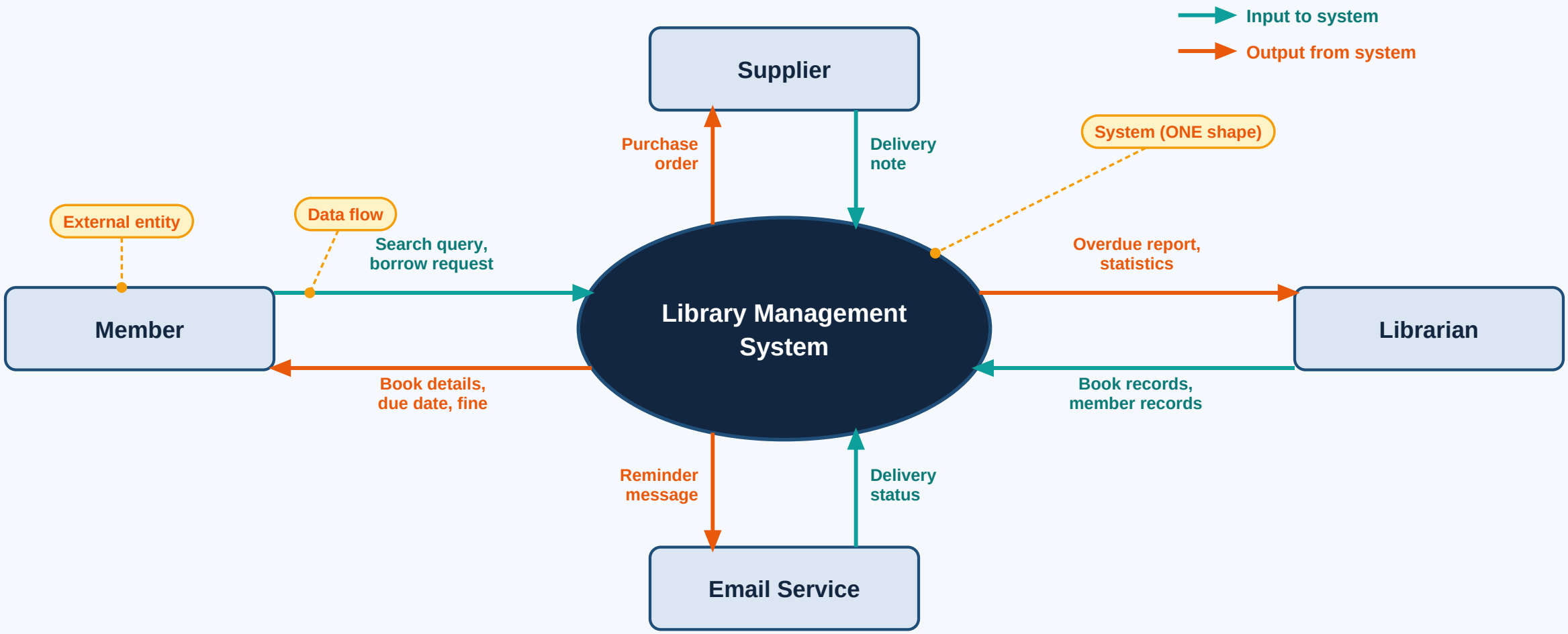
Building the Context Diagram



Step 2: Add the external entities

Who is OUTSIDE and sends or receives data? Member, Librarian, Supplier, Email Service.

Building the Context Diagram



Step 3: Add labeled data flows

Teal = into the system. Orange = out of the system. Every arrow carries named data.

2. UML

Unified Modeling Language

A standard VISUAL notation to model software.

Why?

- Communicate ideas
- Design before coding
- One common language

OMG standard (1997) | ISO standard (2005)

3. Use Case Diagram

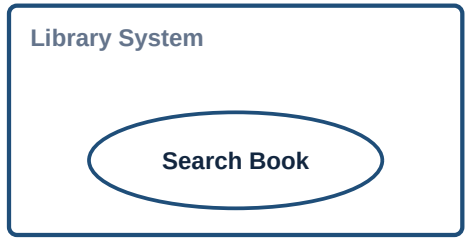
Shows WHAT the system does for its users: Actors + Use Cases. Not HOW.

Actor

Outside the system.
A role, not a name.

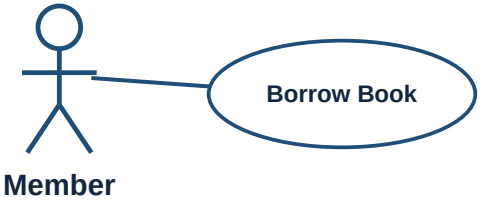
Use Case

One goal.
Verb + Noun.



System Boundary

Inside = use cases.
Outside = actors.



Association

Actor takes part
in the use case.

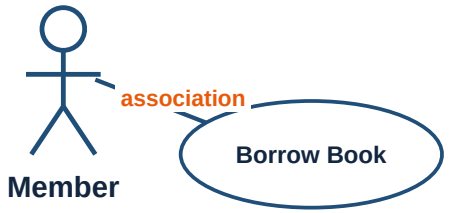
Primary actor

Starts the use case. Drawn on the LEFT.
Example: Member, Librarian

Secondary actor

Supports the system. Drawn on the RIGHT.
Example: Email Service

4. Relationships

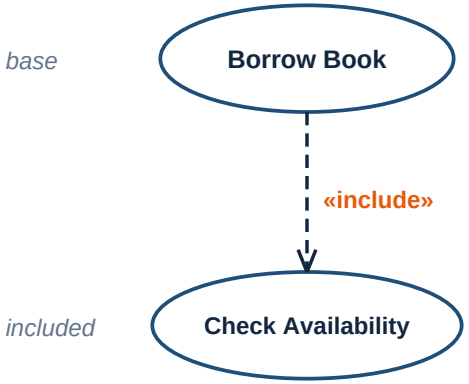


Association

Symbol: solid line

Actor takes part in a use case.

Example
Member - Borrow Book

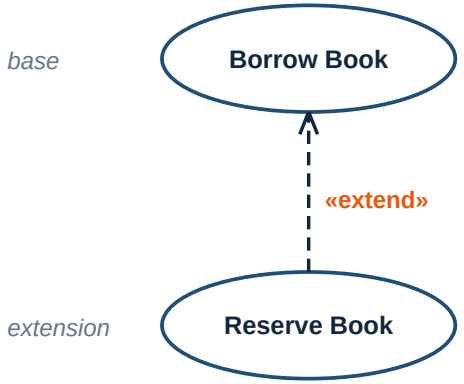


«include»

Symbol: dashed arrow to the INCLUDED

Base ALWAYS runs it. (mandatory)

Example
Borrow Book includes Check Availability

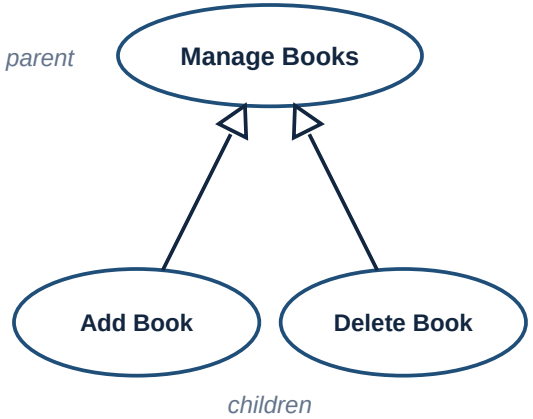


«extend»

Symbol: dashed arrow to the BASE

Runs only IF a condition is true.
(optional)

Example
Reserve Book extends Borrow Book



Generalization

Symbol: hollow triangle to the PARENT

Child = special kind of parent.

Example
Add Book, Delete Book specialize Manage Books

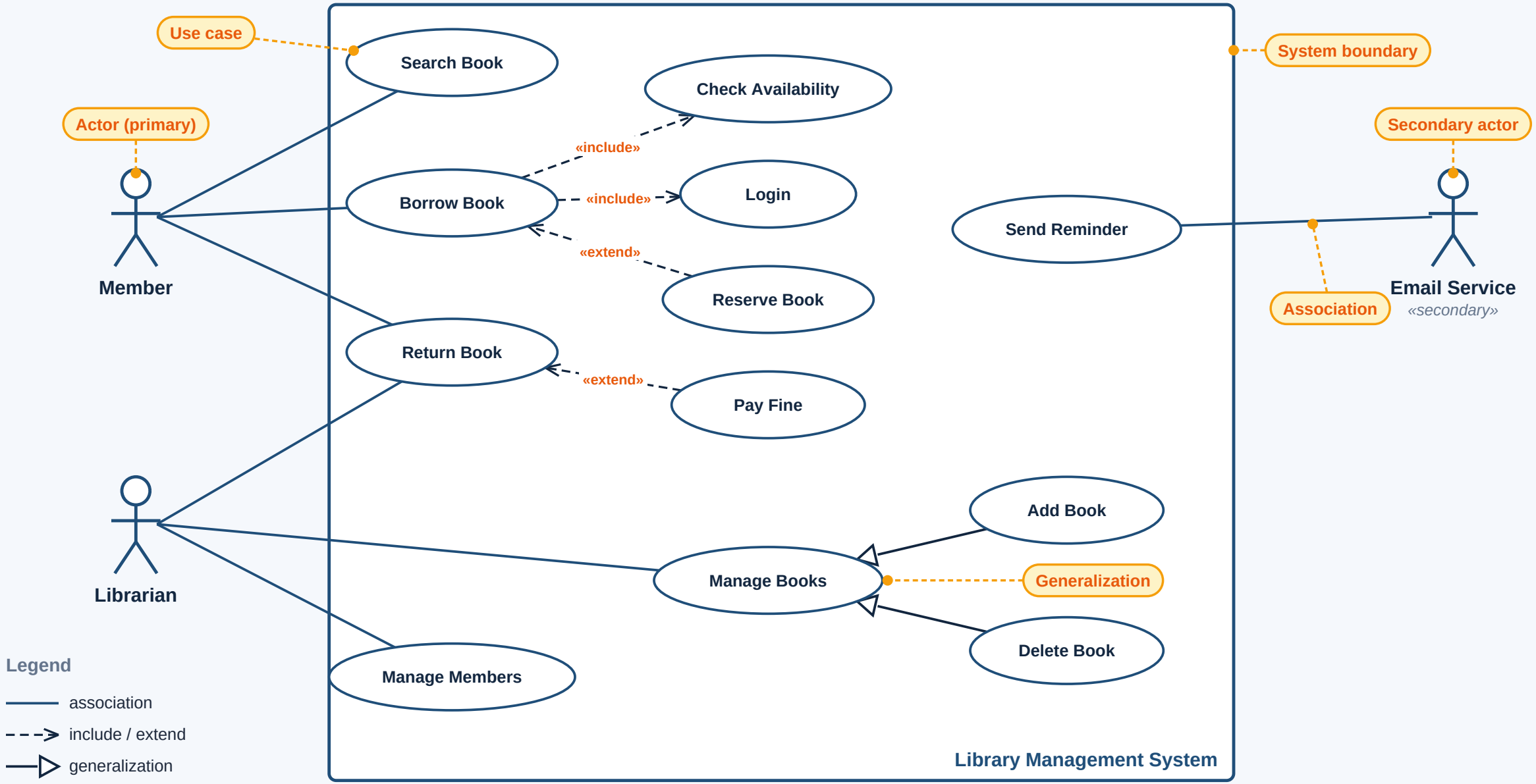
5. Include vs. Extend

	INCLUDE	EXTEND
Meaning	Mandatory part	Optional add-on
Runs	ALWAYS	Only IF condition is true
Arrow points to	The included use case	The base use case
Example	Borrow Book «include» Check Availability	Reserve Book «extend» Borrow Book
Test	Base cannot finish without it	Base works fine without it

Trick

INCLUDE = a must (arrow to the INCLUDED).
EXTEND = an option (arrow to the BASE).

6. Use Case Diagram: Library Management System



7. Use Case Description: "Borrow Book"

Use case	Borrow Book
Actor	Member (primary)
Pre-condition	Member has an account
Main flow	1. Member selects a book and presses Borrow 2. System checks login («include» Login) 3. System checks stock («include» Check Availability) 4. System saves the loan and due date 5. System shows a confirmation
Alternative flow	3a. Book not available: Reserve Book («extend»)
Post-condition	Loan is saved, book status = Borrowed